ENDG 319

Probability, Statistics and Machine Learning

Lecture 8: Linear Functions and Joint Distributions

Transformations • Conditioning • Covariance and correlation

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Learning outcomes for Lecture 8

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FROM ONE RANDOM VARIABLE TO A JOINT PROBABILITY MODEL

1	**Combine**	Find means and variances of linear functions and sample means.
2	**Describe**	Read joint distributions and obtain marginal distributions.
3	**Condition**	Construct conditional distributions and conditional expectations.
4	**Assess dependence**	Use independence, covariance, and correlation appropriately.

> Define the quantity of interest, identify the assumptions, and then choose the rule.

Linear transformations: shift, scale, and reflect

LET Y = AX + B; A AND B ARE CONSTANTS

MEAN

$$\mu_Y = a\mu_X + b$$

Scale the centre, then shift.

VARIANCE

$$\sigma_Y^2 = a^2\sigma_X^2$$

Square the scale factor.

STANDARD DEVIATION

$$\sigma_Y = |a|\sigma_X$$

Use the absolute scale factor.

A constant shift changes the centre—not the variance or standard deviation.

Means of linear combinations: independence is not required

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THE MEAN OF A SUM IS THE SUM OF THE MEANS

Two random variables

$$E[aX + bY] = aE[X] + bE[Y] = a\mu_X + b\mu_Y$$

Any finite linear combination

$$E\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i \mu_{X_i}$$

> Combine the means with the original coefficients, including their signs.

Independence applies to events defined by the variables

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X AND Y CAN EACH DEFINE MANY DIFFERENT EVENTS

$$P(X \in S,\ Y \in T) = P(X \in S)\, P(Y \in T)$$

EVENT FOR X

The value of X falls in the set S.

EVENT FOR Y

The value of Y falls in the set T.

$$P(X_1 \in S_1, \ldots, X_n \in S_n) = \prod_{i=1}^{n} P(X_i \in S_i)$$

Independence concerns the entire joint behaviour, not one convenient probability.

For independent variables, add weighted variances

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SQUARE EVERY COEFFICIENT—EVEN A NEGATIVE ONE

$$\mathrm{Var}\left(\sum_{i=1}^{n} c_i X_i\right) = \sum_{i=1}^{n} c_i^2 \sigma_{X_i}^2$$

SUM

$$\mathrm{Var}(X+Y) = \sigma_X^2 + \sigma_Y^2$$

DIFFERENCE

$$\mathrm{Var}(X-Y) = \sigma_X^2 + \sigma_Y^2$$

Both results on this slide require the stated independence assumption.

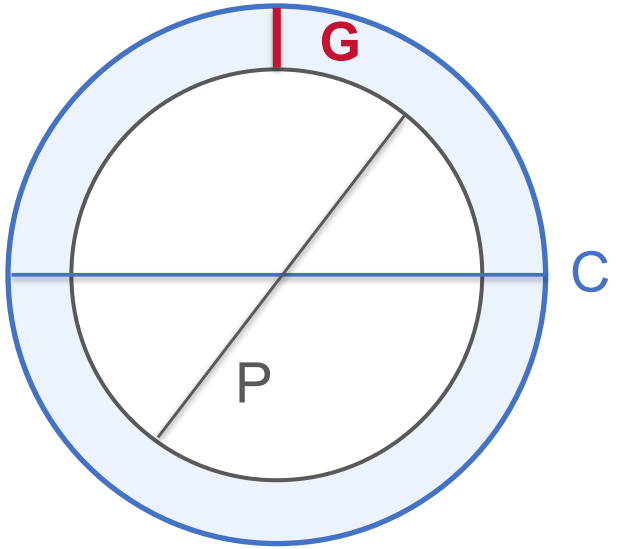
Add variances under independence; do not add standard deviations.

Example 18: model the piston–cylinder clearance

CLEARANCE IS ONE-HALF THE DIFFERENCE IN DIAMETERS

Quantity	Mean (cm)	SD (cm)
Piston P	80.85	0.02
Cylinder C	80.95	0.03

$$G = \frac{C - P}{2} = \frac{1}{2}C - \frac{1}{2}P$$



Schematic—not to scale

Write the physical relationship first: a radial gap is not the full diameter difference.

Example 18: mean and standard deviation of clearance

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USE THE SIGN FOR THE MEAN; SQUARE IT FOR THE VARIANCE

1 Mean clearance

$$\mu_G = \frac{1}{2}(80.95 - 80.85) = 0.050 \text{ cm}$$

2 Variance and standard deviation (independent C and P)

$$\sigma_G^2 = \left(\frac{1}{2}\right)^2 (0.03)^2 + \left(-\frac{1}{2}\right)^2 (0.02)^2 = 0.000325 \text{ cm}^2$$

$$\sigma_G = \sqrt{0.000325} \approx 0.01803 \text{ cm}$$

Mean clearance = 0.050 cm; standard deviation ≈ 0.01803 cm.

A simple random sample and the i.i.d. model

STATE THE SAMPLING ASSUMPTIONS BEFORE USING THE FORMULAS

INDEPENDENT

Observations do not provide information about one another.

IDENTICALLY DISTRIBUTED

Every observation follows the same population distribution.

Together: independent and identically distributed (i.i.d.).

Finite population, without replacement: independence is an approximation when the sampling fraction is negligible.

> Use the sample-mean variance formula only under its sampling model.

The sample mean is itself a random variable

FOR I.I.D. OBSERVATIONS WITH MEAN M AND VARIANCE Σ²

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu$$

$$\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 = \frac{\sigma^2}{n}$$

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

Averaging preserves the population mean and reduces variability across samples.

Example 19: average fill volume in a case

24 BOTTLES; MEAN 2.013 L AND STANDARD DEVIATION 0.005 L

Quantity	Mean (L)	Standard deviation (L)
One bottle	2.013	0.005
Average of 24 bottles	2.013	0.001021

$$\mu_{\overline{X}} = \mu = 2.013 \text{ L}$$

$$\sigma_{\overline{X}} = \frac{0.005}{\sqrt{24}} \approx 0.001021 \text{ L}$$

The case average varies less than a single bottle; its mean remains 2.013 L.

Checkpoint: choose the correct moment rule

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DISCUSS THE REASONING BEFORE CALCULATING

A	**Add a constant**	Which of the mean, variance, and SD changes for X + b?
B	**Take a difference**	For independent X and Y, do their variances add or subtract in X − Y?
C	**Average more values**	Under the same i.i.d. model, what happens to the SD of the mean when n quadruples?

For each answer, name the rule and its assumptions.

Checkpoint: answers and explanations

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RETURN TO THE FORMULA—NOT THE WORDING ALONE

A	**Only the mean**	$E[X + b] = E[X] + b$. The spread is unchanged.
B	**The variances add**	$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$, under independence.
C	**The SD is halved**	$\sigma / \sqrt{(4n)} = (1/2)(\sigma / \sqrt{n})$. The mean remains μ.

A sign changes a mean; a squared coefficient determines its variance contribution.

A joint PMF describes two discrete variables together

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ONE PROBABILITY FOR EACH POSSIBLE PAIR (X, Y)

$$p(x, y) = P(X = x,\ Y = y)$$

EACH CELL

$p(x, y) \geq 0$

ALL CELLS

$$\sum_x \sum_y p(x, y) = 1$$

An event is a collection of pairs; add the probabilities of its cells.

Joint means “together”: the elementary numerical outcome is the pair (x, y).

Marginal PMFs: sum over the other variable

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THE ROW AND COLUMN TOTALS EACH FORM A ONE-VARIABLE PMF

X \ Y	y_1	y_2	Row total
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	$p_X(x_1)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	$p_X(x_2)$
Column total	$p_Y(y_1)$	$p_Y(y_2)$	1

$$p_X(x) = \sum_y p(x, y)$$

$$p_Y(y) = \sum_x p(x, y)$$

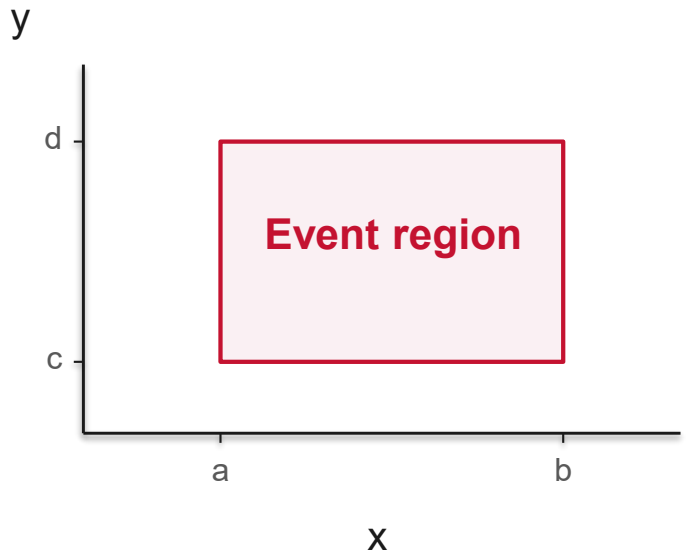
Marginalize by summing out the variable you are not retaining.

A joint PDF assigns probability to a region

FOR JOINTLY CONTINUOUS X AND Y, INTEGRATE OVER THE EVENT REGION

$$P(a \leq X \leq b,\ c \leq Y \leq d)$$

$$= \int_a^b \int_c^d f(x,y)\, dy\, dx$$



$$f(x,y) \geq 0, \qquad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dy\, dx = 1$$

Draw the event region before writing the limits of integration.

Marginal PDFs: integrate over the other variable

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KEEP ONE COORDINATE FIXED AND INTEGRATE OUT THE OTHER

Retain X

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

Retain Y

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

A marginal density describes one variable without conditioning on the other.

Example 20: washer thickness and hole diameter

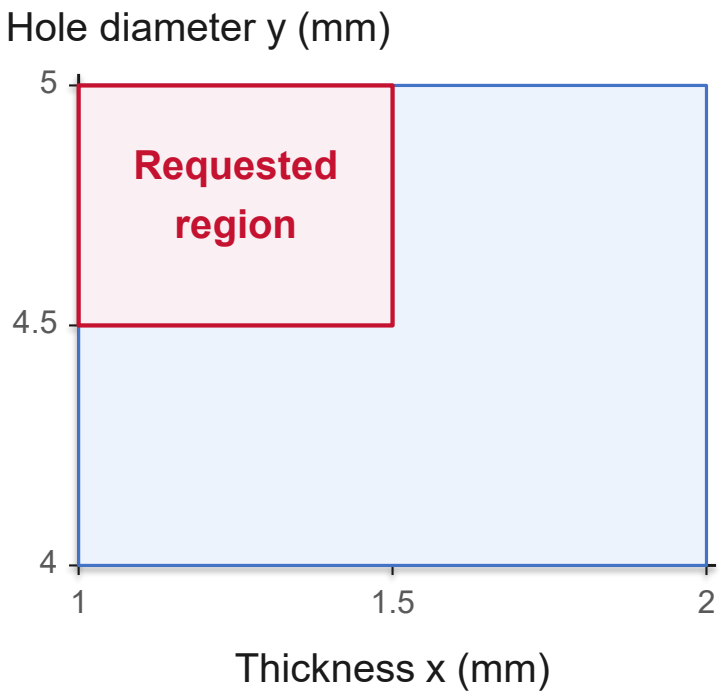
ONE MODEL USED FOR JOINT, MARGINAL, AND CONDITIONAL CALCULATIONS

X: thickness (mm)

Y: hole diameter (mm)

$$f(x,y) = \begin{cases} \dfrac{x+y}{6}, & 1 \le x \le 2,\ 4 \le y \le 5, \\ 0, & \text{otherwise.} \end{cases}$$

Find a region probability and the two marginal densities.



The support tells you where the density is nonzero—not that it is uniform.

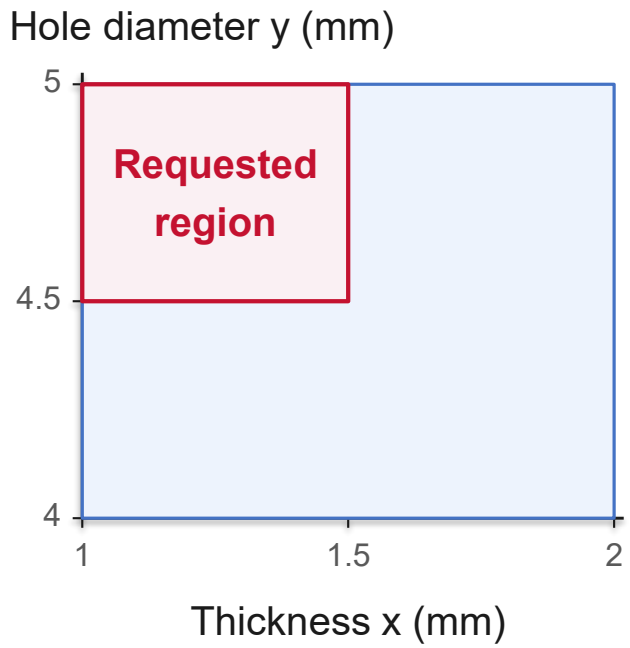
Example 20: integrate over the requested rectangle

THICKNESS 1.0–1.5 MM AND HOLE DIAMETER 4.5–5.0 MM

$$P = \int_{1}^{1.5} \int_{4.5}^{5} \frac{x+y}{6}\, dy\, dx$$

$$= \int_{1}^{1.5} \frac{0.5x + 2.375}{6}\, dx$$

$$= \left[\frac{0.25x^2 + 2.375x}{6} \right]_{1}^{1.5} = \frac{1}{4}$$



P(1 ≤ X ≤ 1.5, 4.5 ≤ Y ≤ 5) = 0.25.

Example 20: the two marginal densities

INTEGRATE OVER THE FULL SUPPORT OF THE OTHER VARIABLE

THICKNESS X

$$f_X(x) = \int_4^5 \frac{x+y}{6}\,dy$$

$$f_X(x) = \begin{cases} (x+4.5)/6, & 1 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

HOLE DIAMETER Y

$$f_Y(y) = \int_1^2 \frac{x+y}{6}\,dx$$

$$f_Y(y) = \begin{cases} (y+1.5)/6, & 4 \le y \le 5, \\ 0, & \text{otherwise.} \end{cases}$$

A complete density includes both its expression and its support.

The same framework extends to more than two variables

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REPLACE A PAIR OF VALUES WITH A VECTOR OF VALUES

Jointly discrete

$$p(x_1, \ldots, x_n) = P(X_1 = x_1, \ldots, X_n = x_n)$$

Jointly continuous

$$P(a_1 \leq X_1 \leq b_1, \ldots, a_n \leq X_n \leq b_n)$$

$$= \int_{a_n}^{b_n} \cdots \int_{a_1}^{b_1} f(x_1, \ldots, x_n)\, dx_1 \cdots dx_n$$

A joint distribution records which combinations of values can occur and how likely they are.

Expected values of functions: transform, then average

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YOU CAN COMPUTE E[H(X)] FROM THE DISTRIBUTION OF X

DISCRETE X

$$E[h(X)] = \sum_x h(x)p(x)$$

CONTINUOUS X

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)f(x)\,dx$$

In general, E[h(X)] is not h(E[X]); use the probability weights.

Expected values of functions of two variables

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USE THE JOINT DISTRIBUTION AS THE WEIGHTING FUNCTION

JOINTLY DISCRETE

$$E[h(X,Y)] = \sum_x \sum_y h(x,y)p(x,y)$$

JOINTLY CONTINUOUS

$$E[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y)f(x,y)\,dy\,dx$$

Use p(x, y) or f(x, y)—not a product of marginals unless independence is justified.

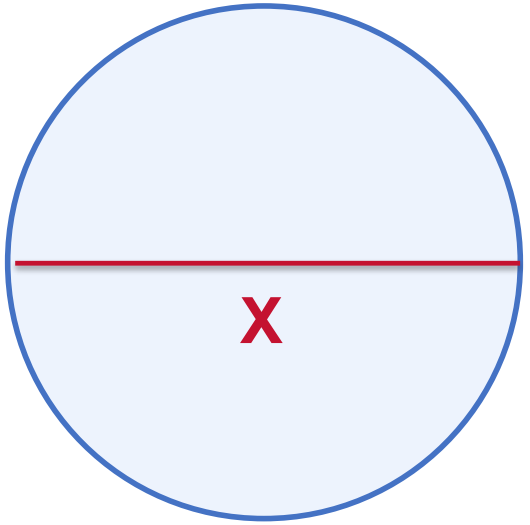
Example 21: mean area of a cylinder bore

THE AREA DEPENDS ON THE SQUARE OF THE BORE DIAMETER

X is the bore diameter (mm).

$$f(x) = \begin{cases} 10, & 80.5 < x < 80.6, \\ 0, & \text{otherwise.} \end{cases}$$

$$A = \frac{\pi X^2}{4} \qquad \text{Find } E[A].$$



Bore cross-section

The question is E[πX²/4], not simply the area evaluated at the mean diameter.

Example 21: compute the expectation of the area

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MULTIPLY THE FUNCTION BY THE DENSITY AND INTEGRATE

$$E[A] = \int_{80.5}^{80.6} \frac{\pi x^2}{4} (10)\, dx$$

$$= \frac{10\pi}{12}\left(80.6^3 - 80.5^3\right)$$

$$= \frac{1946491\pi}{1200} \approx 5095.90 \text{ mm}^2$$

Use E[X²], not (E[X])², in the area calculation.

Mean bore area ≈ 5095.90 mm², obtained by averaging the squared diameter.

Conditional PMFs: fix x and renormalize

DIVIDE THE JOINT PROBABILITY BY THE MARGINAL FOR THE GIVEN VARIABLE

$$p_{Y|X}(y \mid x) = \frac{p(x,y)}{p_X(x)}, \qquad p_X(x) > 0$$

FIX

Hold X at the stated value x.

DIVIDE

Divide by the X row total.

CHECK

Conditional probabilities sum to 1.

The denominator belongs to the variable after the conditioning bar.

Conditional PDFs: a normalized slice of the joint density

FIX X; THE RESULTING DENSITY IS A FUNCTION OF Y

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}, \qquad f_X(x) > 0$$

$$\int_{-\infty}^{\infty} f_{Y|X}(y \mid x)\, dy = 1$$

Conditional probabilities are areas under this one-variable density.

After conditioning on X = x, integrate with respect to Y—not X.

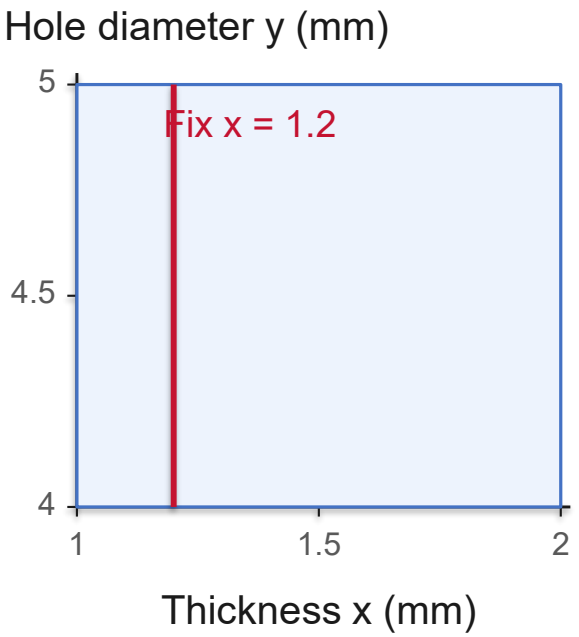
Example 20: condition on a thickness of 1.2 mm

FIND P(Y ≤ 4.8 | X = 1.2)

$$f_X(1.2) = \frac{1.2 + 4.5}{6} = 0.95$$

$$f_{Y|X}(y \mid 1.2) = \frac{1.2 + y}{5.7}, \quad 4 \le y \le 5$$

$$P = \int_4^{4.8} \frac{1.2 + y}{5.7}\, dy = \frac{224}{285} \approx 0.785965$$



P(Y ≤ 4.8 | X = 1.2) ≈ 0.785965, or 78.60%.

Conditional expectation: average within the conditional law

THE WEIGHTING DISTRIBUTION HAS CHANGED; THE AVERAGING IDEA HAS NOT

DISCRETE

$$E[Y \mid X = x] = \sum_{y} y\, p_{Y|X}(y \mid x)$$

CONTINUOUS

$$E[Y \mid X = x] = \int_{-\infty}^{\infty} y\, f_{Y|X}(y \mid x)\, dy$$

Use the conditional distribution—not the marginal—to compute a conditional mean.

Example 20: the conditional mean hole diameter

FIND E[Y | X = 1.2]

$$E[Y \mid X = 1.2] = \int_4^5 y \, \frac{1.2 + y}{5.7} \, dy$$

$$= \frac{1}{5.7} \left[0.6y^2 + \frac{y^3}{3} \right]_4^5$$

$$= \frac{772}{171} \approx 4.51462 \text{ mm}$$

The conditional mean is approximately 4.51462 mm—a value, not a probability.

Independence means the joint distribution factors

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A CONDITION ON THE WHOLE DISTRIBUTION

JOINTLY DISCRETE

$$p(x, y) = p_X(x)p_Y(y)$$

JOINTLY CONTINUOUS

$$f(x, y) = f_X(x)f_Y(y)$$

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p_{X_i}(x_i)$$

$$f(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i)$$

A matching cell or point is not enough; factorization is a distribution-wide condition.

Under independence, conditioning changes nothing

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THE CONDITIONAL DISTRIBUTION EQUALS THE MARGINAL DISTRIBUTION

$$p_{Y|X}(y \mid x) = \frac{p_X(x)p_Y(y)}{p_X(x)} = p_Y(y)$$

For a positive marginal probability $p_X(x)$.

$$f_{Y|X}(y \mid x) = \frac{f_X(x)f_Y(y)}{f_X(x)} = f_Y(y)$$

For a positive marginal density $f_X(x)$.

Independent variables: knowing X does not update the distribution of Y.

Example 20: are thickness and hole diameter independent?

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COMPARE THE JOINT DENSITY WITH THE PRODUCT OF ITS MARGINALS

$$f_X(x)f_Y(y) = \frac{(x+4.5)(y+1.5)}{36}$$

$$f(x,y) - f_X(x)f_Y(y) = -\frac{(x-1.5)(y-4.5)}{36}$$

This difference is not zero throughout the support.

A rectangular support alone does not establish independence.

The washer variables are dependent: their joint density does not factor into the marginals.

Covariance measures joint deviations from the means

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AVERAGE THE PRODUCT OF THE TWO CENTRED VARIABLES

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

$$= E[XY] - \mu_X\mu_Y$$

POSITIVE CONTRIBUTION

Deviations have the same sign.

NEGATIVE CONTRIBUTION

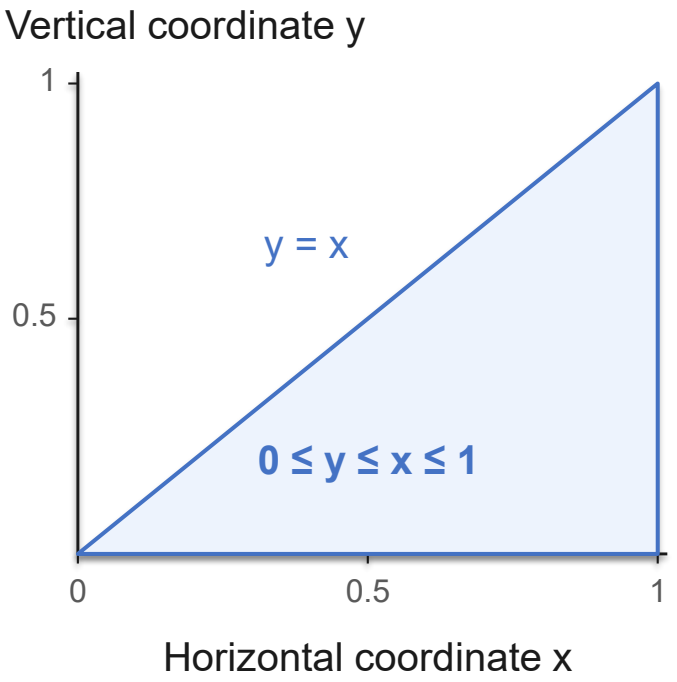
Deviations have opposite signs.

Covariance uses the joint moment E[XY], not just the two marginal means.

Example 22: position of a mobile computer

THE SUPPORT IS A TRIANGLE—NOT THE FULL UNIT SQUARE

$$f(x,y) = \begin{cases} 8xy, & 0 \le y \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$



Find Cov(X, Y).

We need E[X], E[Y], and E[XY].

Use 0 ≤ y ≤ x ≤ 1; the upper limit for y depends on x.

Example 22: compute the three moments for covariance

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EVERY INTEGRAL USES THE SAME TRIANGULAR SUPPORT

$$E[X] = \int_0^1 \int_0^x x(8xy)\,dy\,dx = \frac{4}{5}$$

$$E[Y] = \int_0^1 \int_0^x y(8xy)\,dy\,dx = \frac{8}{15}$$

$$E[XY] = \int_0^1 \int_0^x xy(8xy)\,dy\,dx = \frac{4}{9}$$

$$\mathrm{Cov}(X,Y) = \frac{4}{9} - \frac{4}{5}\frac{8}{15} = \frac{4}{225}$$

Cov(X, Y) = 4/225 ≈ 0.01778; retain exact fractions until the final step.

Correlation standardizes covariance

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DIVIDE BY THE PRODUCT OF THE TWO STANDARD DEVIATIONS

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$-1 \leq \rho_{X,Y} \leq 1$$

Assume finite, strictly positive standard deviations.

Correlation is dimensionless; its denominator is $\sigma_X\sigma_Y$, not $\sigma_X^2\sigma_Y^2$.

Example 22: second moments and variances

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COMPUTE THE STANDARD DEVIATIONS NEEDED FOR CORRELATION

$$E[X^2] = \int_0^1 \int_0^x x^2(8xy)\,dy\,dx = \frac{2}{3}$$

$$\sigma_X^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75}$$

$$E[Y^2] = \int_0^1 \int_0^x y^2(8xy)\,dy\,dx = \frac{1}{3}$$

$$\sigma_Y^2 = \frac{1}{3} - \left(\frac{8}{15}\right)^2 = \frac{11}{225}$$

Variance = second moment − squared mean; there is no sample n − 1 divisor here.

Example 22: the correlation coefficient

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COMBINE THE EXACT COVARIANCE AND VARIANCES

Cov(X, Y)	Var(X)	Var(Y)
4/225	2/75	11/225

$$\rho_{X,Y} = \frac{4/225}{\sqrt{2/75}\sqrt{11/225}}$$

$$= \frac{4}{\sqrt{66}} \approx 0.4924$$

$\rho_{X,Y} = 4/\sqrt{66} \approx 0.4924$: positive linear association in the stated model.

Uncorrelated does not necessarily mean independent

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KEEP THE LOGICAL DIRECTION CLEAR

$$\text{Independence} \implies \text{Cov}(X, Y) = 0$$

$$\text{Cov}(X, Y) = 0 \not\implies \text{independence}$$

When σ_X and σ_Y are positive, zero covariance is equivalent to zero correlation.

Zero correlation rules out linear association—not every form of dependence.

General linear combinations: keep the covariance terms

THE MEAN RULE IS UNCHANGED; THE VARIANCE RULE NEEDS CROSS-TERMS

$$Z = \sum_{i=1}^{n} c_i X_i, \qquad E[Z] = \sum_{i=1}^{n} c_i \mu_{X_i}$$

$$\mathrm{Var}(Z) = \sum_{i=1}^{n} c_i^2 \sigma_{X_i}^2 + 2 \sum_{i<j} c_i c_j \,\mathrm{Cov}(X_i, X_j)$$

Each distinct pair contributes once inside the second sum.

Without a justified zero-covariance condition, the cross-terms cannot be discarded.

Sums and differences: what covariance changes

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SPECIALIZE THE GENERAL RULE TO TWO RANDOM VARIABLES

$$\mathrm{Var}(X+Y) = \sigma_X^2 + \sigma_Y^2 + 2\,\mathrm{Cov}(X,Y)$$

$$\mathrm{Var}(X-Y) = \sigma_X^2 + \sigma_Y^2 - 2\,\mathrm{Cov}(X,Y)$$

Independent X and Y: Cov(X, Y) = 0, so both reduce to $\sigma_X^2 + \sigma_Y^2$.

The minus sign in a difference affects the covariance term—not the two individual variances.

Example 22: translate the travel path into a variable

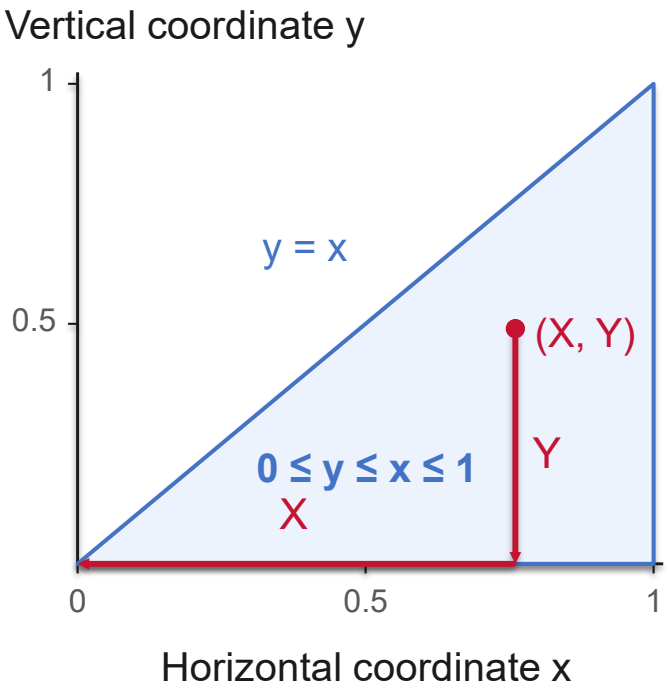
MOVE VERTICALLY TO (X, 0), THEN HORIZONTALLY TO THE ORIGIN

Vertical segment: distance Y

Horizontal segment: distance X

$$D = X + Y$$

Find the mean and variance of D under the original joint model.



For the specified two-segment route, D = X + Y—not $\sqrt{(X^2 + Y^2)}$.

Example 22: mean and variance of travel distance

THE COORDINATES ARE DEPENDENT, SO RETAIN THEIR COVARIANCE

$$E[D] = \frac{4}{5} + \frac{8}{15} = \frac{4}{3}$$

$$\mathrm{Var}(D) = \frac{2}{75} + \frac{11}{225} + 2\left(\frac{4}{225}\right)$$

$$= \frac{25}{225} = \frac{1}{9}$$

Mean: coordinate units. Variance: squared coordinate units.

E[D] = 4/3 and Var(D) = 1/9; the covariance contribution is essential.

Final checkpoint: joint, marginal, or conditional?

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IDENTIFY THE OPERATION AND THE ASSUMPTION

A	**Marginal density**	To obtain $f_X(x)$ from $f(x, y)$, which variable do you integrate out?
B	**Conditional density**	For the density of Y given X = x, which marginal is the denominator?
C	**Zero correlation**	Does $\rho_{X,Y} = 0$ establish independence?
D	**Variance of a sum**	What additional term is needed when Cov(X, Y) is nonzero?

> Name the target distribution before choosing a formula.

Final checkpoint: answers

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EXPLAIN WHY EACH OPERATION MATCHES THE QUESTION

A	**Integrate out Y**	Keep x fixed; integrate the joint density with respect to y.
B	**Divide by $f_X(x)$**	The denominator corresponds to the given variable X.
C	**No**	Uncorrelated variables can still be dependent.
D	**Add 2Cov(X, Y)**	Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y).

Mean rules are linear; variance rules must account for dependence.

Lecture 8: key takeaways

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A SINGLE WORKFLOW CONNECTS ALL OF TODAY'S METHODS

Linear functions	Means combine linearly; variances use squared coefficients.
Sample means	Under the i.i.d. model, the SD of the mean is $\sigma/\sqrt{n}$.
Joint and marginal	Sum or integrate out the variable you do not retain.
Conditioning	Fix the given value and normalize by its marginal.
Dependence	Independence is stronger than zero correlation.
General variance	Retain covariance terms unless their absence is justified.

Define the quantity → identify the distribution and assumptions → calculate → interpret.